

■ Cash & Bank Management Module & Business Cycles

D365FO Middleware Backend — Cash Receipts, Disbursements & FX Balancing

Target Audience: Treasury, Cash Office, AR/AP Collections, Software Engineers.

1. Business Domain Overview

The **Cash & Bank Management** module handles all treasury and liquidity transactions:

- **Cash-In (Collections & Receipts):** Recording customer payments, bank deposits, down payments, and wire receipts for freight and fleet trucking services.
- **Cash-Out (Disbursements & Advances):** Recording vendor payments, bank transfers, petty cash disbursements, and driver operational cash advances.
- **FX Exchange Rate Balancing:** Adjusting foreign currency receipts/payments against original invoice exchange rates to record realized foreign exchange gains or losses.
- **Custody Settlement Routing:** Isolating custody advance settlements and routing them to dedicated clearing journals.

2. Cash Entry Processors & Business Scope

Processor Class Name	EntryProcessorTypes	D365FO Target Entity	Real-World Business Context
CashInFreightEntryProcessor	CashInFreight	CustPaymentJournalHeaders / Lines	Customer Payment Receipts (Freight)
CashInTruckingEntryProcessor	CashInTrucking	CustPaymentJournalHeaders / Lines	Customer Payment Receipts (Trucking)
CashOutFreightEntryProcessor	CashOutFreight	Disbursement Journal	Vendor & Cash Disbursements (Freight)
CashOutTruckingEntryProcessor	CashOutTrucking	Disbursement Journal	Vendor & Cash Disbursements (Trucking)

3. Financial & Accounting Impact in D365FO

■ Cash-In (Customer Collection Journal - Cust - Pay):

- **DEBIT:** Bank / Cash Main Account (e.g., **110201** CIB EGP Bank or **110202** USD Bank)
- **CREDIT:** Customer Subledger Account (**CustAccount** e.g. **CUST-00192**)

- **DEBIT/CREDIT (FX Difference):** Realized FX Gain/Loss Account (e.g., **710101** / **810101**) if invoice FX rate differs from payment date FX rate.

■ **Cash-Out (Vendor Disbursement Journal):**

- **DEBIT:** Vendor Subledger Account (**VendAccount** e.g. **VEND-00412**) or Expense/Custody Account
- **CREDIT:** Bank / Cash Main Account

4. End-to-End Cash Business Cycle Flow

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[Bank Statements / Cash Office Receipt Vouchers]
■
▼
[API Endpoint: Upload Cash Receipts Excel]
■
▼
[CashInFreightEntryProcessor]
(Parses Bank Accounts, Customer IDs, FX Differences)
■
▼
[DataBatch Status: PENDING (Max 1000 Lines)]
■
▼
[BullMQ Queue: dfo-customer-payment-journal-queue]
■
▼
[CustomerPaymentPostingStrategy]
■
▼
[Post OData to D365FO CustPaymentJournalHeaders]
■
▼
[D365FO Posts Customer Payment Journal]
(Clears Open Invoice & Balances Bank/Customer Subledgers)
■
▼
[Batch Status Set to COMPLETED]

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Detailed Step-by-Step Cycle Execution:

1. **Cash Sheet Upload:** Cashier or treasury accountant uploads daily collection logs containing bank account codes, customer codes, payment amounts, and invoice reference numbers.
2. **Custody Split:** Any lines marked as driver custody settlements are separated and routed to the custody settlement workflow.
3. **FX Calculation:** If a customer pays in USD for an invoice originally issued in USD but recorded in EGP, the system fetches transaction date exchange rates to balance net GL amounts.
4. **Queue Posting:** Dispatched to **dfo-customer-payment-journal-queue**.
5. **OData Execution:** Posts header and line records into D365FO customer payment journal.
6. **Subledger Settlement:** D365FO applies the cash receipt against open customer invoices, updating customer balance.